

## **Introduction**

As a part-time React web developer for 2T-Security, I have the opportunity to work with a company that provides expert consultancy solutions in areas such as risk management, security architecture and security monitoring. By leveraging their expertise, 2T-Security help clients solve their IT security and information assurance challenges and achieve their goals.

2T-Security are a small sized, Remote-based company with around 10-15 employees with a development team of 4 members excluding me, the development team mainly work on the businesses web applications tools; Time Tracking, RiskTree and the Company Website. They are a remote based company meaning that all employees carry out most of their work from home so online collaborative tools such as Microsoft Teams, Jira Work Management and GitHub are key to their business processes.

2T Security was setup in 2013 by 3 ex IBM employees who didn't enjoy the corporate working culture at large companies, the culture they wanted to breed at 2T Security is one where employees felt comfortable, respected and heard. It was important that business decisions were driven only by what was right from a moral and security standpoint instead of being driven by financially motivated goals.

## **Software Development Practices at 2T Security**

My role at the company is focused on maintaining the company website by addressing any bugs/problems and updating the website by adding new content and features. As I am a full-time student 2T Security have taken this into consideration and setup a work-life structure where I am able to work whenever I want, no matter what time or day.

The development team use the agile development workflow strategy to carry out work on the business internal tools, Agile development is a set of principles that guide the development of software in a flexible and iterative manner. It emphasizes collaboration, continuous delivery, and rapid iteration to quickly deliver value to customers. One key way 2T-Security implements agile working and encourages collaboration is the use of daily stand-up meetings, these meetings are usually short 10-15 minute meetings in the morning which provide a regular opportunity for the development team to share what they are working on and any challenges or roadblocks they are facing, the meeting minutes are regularly posted on 'dev chat' which is a group on Microsoft Teams just for the developers. This helps ensure that everyone is on the same page and can work together more effectively.

## **Daily Stand-up Meetings**

As a part-time worker I wasn't required to attend every daily stand-up meeting but attended as many as I could, I found these meetings to be extremely useful because I was able to get input and ideas from other developers on any problems I was facing, and more often than not these solutions would help me overcome the roadblocks I was facing with the ticket I was completing. It also gives me the opportunity to learn about what other people are doing and how they are doing it, this has been useful to me during my university work because I am able to use their solutions and ideas to my own problems, for example in Software Quality Assurance for the group coursework me and my group were brainstorming the best ways to be able to track testing. As I had attended the daily stand up meeting where one of the developers had talked about how he was automating testing in git using GitHub actions I came up with the idea to implement an automated pipeline for my group so

that we were all able to track testing metrics easily and not rely on the group's developers to manually take notes of the testing results for our report.

Daily stand-up meetings were something I didn't realize or appreciate before working in industry, I had learnt about them during the Introduction to Software Engineering COMP1003 module but never put much importance to it, now I have seen it used practically in a professional environment I appreciate how beneficial it can be for personal development and the development and unity of the team.

### **Feature Branching**

When collaborating and working together 2T Security use GitHub along with Jira to manage their workload and delegate tasks.

2T Security use the Feature branch workflow strategy which is that all feature development takes place in a dedicated branch instead of the main branch. By working on separate branches, team members can work on their own features without worrying about conflicting with the work of others. This makes it easier for team members to collaborate and review each other's work. When a feature is complete, it is merged back into the main branch. This makes code easier to integrate and improves code quality when merging into the main codebase and helps to ensure that the codebase remains stable. As 2T Security have a small development team it is crucial the team is able to work with flexibility, feature branching also enhances flexibility because it allows developers to work on multiple features at the same time and allows them to respond to changing priorities or requirements with ease.

The Feature branching strategy is also perfect for me and remote part-time working because of the flexibility it allows. Each ticket in the 2T Security Jira board represents a single new feature or bug to be fixed, tickets are tagged with an estimated complete time calculated by the lead developer in dev team, this along with my day rate is used to calculate how much I would get paid. For example if the ticket has an estimated complete time of 1 day, regardless of how long it took me to complete the ticket I would be paid for 1 day's worth of work. Tickets are also labelled based on their urgency with the majority of tickets being non-urgent tickets with no time constraints. As the work I do is not dependent on other developers and tickets have no time constraints it allows me to work flexibly and whenever I want, no matter the day or time.

I have enjoyed this style of working because of how flexible it is, it does not put additional stress on top of university work and allows me to learn and develop my skills at my own pace. Working with no strict time constraints allows me to produce quality work and gives me time to fully understand the solution I have implemented. It also allows me time to improve or experiment with better more efficient solutions.

### **Rubber Duck**

The Lead developer set up a "Rubber Duck" channel to help developers when debugging their code; Rubber duck debugging is a software development technique in which a developer explains their code, step by step, to a rubber duck or other inanimate object. The idea is that by talking through the code, the developer may find the root cause of a bug or issue. This technique is something new that was introduced to me by 2T Security and is used across the company by developers. I surprisingly found this to be an extremely effective way of getting over a roadblock or identifying problems when debugging code. It forced me as the developer to think through my code and articulate my thought process, it allowed me to "hear" my own explanation and see if it makes sense which more often

than not identified inconsistencies and gaps in my understanding. The channel also was a helpful way to document the code for future reference. By explaining the code into the rubber duck channel, it effectively created a "narrative" of the work I was completing which can be useful for others to reference later. The Rubber Duck technique is one I will carry on using in my professional career and is again one that I only fully appreciated once I used it in practice in a professional environment.

### **CI/CD Pipeline**

Recently 2T Security have introduced some automation tools for their web applications which has numerous benefits for their software development cycle. Automation tools increase speed and efficiency when it comes to getting code ready for live deployment, tests can be run much faster than manual tests, which can save time and resources. Automation also improves accuracy and coverage as tests are less prone to human error and tests can be written to cover a wide range of scenarios, which can help ensure that the software is thoroughly tested and working as intended. The repository for the company website uses the Cypress testing framework; Cypress is a JavaScript-based testing framework that is easy to setup and designed for testing web applications. It was important to 2T that the tool was developer-friendly, easy to integrate and use as they do not have someone on their team with DevOps experience. Installation of the tool is simple and requires no dependencies, extra downloads or changes to the existing code. The tool was implemented by a year in industry student last year who installed and acted as the quality assurance test engineer who's responsibility was to write unit tests for the codebase and ensure usability of new or existing software before it goes live. He has now left and no new QA tester has been recruited which means the automatic unit test cases are not being maintained or updated at the moment, I suggested for the mean time while they look for new QA tester developers could write their own unit tests to test the code. It was then explained to me why it is generally considered a bad practice for developers to write their own unit tests, unit tests should be written by someone other than the person who wrote the code being tested. This helps to ensure that the tests are objective and unbiased, and that they are not influenced by the developer's own assumptions about how the code should behave. It is also generally a good idea to have a separate team or individual responsible for writing and maintaining unit tests. This helps to ensure that the tests are thorough and comprehensive, and that they are kept up to date as the codebase evolves.

2T Security are a company that respect all their employees equally no matter their experience, as a part-time Junior Developer I feel 2T always trusts and respects the solutions and design choices I make. Most tickets available for me are open ended and up to my discretion on how I would like to implement them, one ticket I had recently completed was to create the careers page for the website that would list the available opportunities 2T Security are currently offering. The ticket included the text they would like to include such as the job description for each role but no UI design specification which meant I had the freedom and was trusted to design and implement the page myself. I am supported by the lead developer who I am able to message on Teams Monday to Friday during normal working hours, I regularly ask for feedback about any changes that they would like. Once a ticket has been completed it's my responsibility to create a pull request and resolve any merge conflicts that may arise, the lead developer will then review the code and make comments if needed.

I appreciate the responsibility and trust 2T Security give to me, I feel if I was working for a larger company I wouldn't be able to have the freedom I have at 2T and the work I would be doing wouldn't be as impactful; In a large company I would be probably feel like a small cog in a large machine which I feel wouldn't give me as much satisfaction.